

IMO(35th) – 1994

First Day

1. Let m and n be positive integers. Let $a_1, a_2, \dots, a_m$ be distinct elements of $\{1, 2, \dots, n\}$ such that whenever $a_i + a_j \leq n$ for some $i, j, 1 \leq i \leq j \leq m$, there exist $k, 1 \leq k \leq m$, with $a_i + a_j = a_k$. Prove that $\frac{a_1 + a_2 + \dots + a_m}{m} \geq \frac{n+1}{2}$.
2. ABC is an isosceles triangle with $AB = AC$. Suppose that:
 - (a) M is the midpoint of BC and O is the point on the line AM such that OB is perpendicular to AB.
 - (b) Q is an arbitrary point on the segment BC different from B and C.
 - (c) E lies on the line AB and F lies on the line AC such that E, Q, F are distinct and collinear.
 Prove that OQ is perpendicular to EF if and only if $QE = QF$.
3. For any positive integer k , let $f(k)$ be the number of elements in the set $\{k+1, k+2, \dots, 2k\}$ whose base 2 representation has precisely three 1s.
 - (a) Prove that, for each positive integer m , there exist at least one positive integer k such that $f(k) = m$.
 - (b) Determine all positive integer m for which there exist exactly one k with $f(k) = m$.

Second Day

4. Determine all ordered pairs (m, n) of positive integers such that $\frac{n^3 + 1}{mn - 1}$ is an integer.
5. Let S be the set of real number strictly greater than -1 . Find all functions $f : S \rightarrow S$ satisfying the two conditions:
 - (a) $f(x + f(y) + xf(y)) = y + f(x) + yf(x)$ for all x and y in S ;
 - (b) $\frac{f(x)}{x}$ is strictly increasing on each of the intervals $-1 < x < 0$ and $0 < x$.
6. Show that there exist a set A of positive integers with the following property: For any infinite set S of primes there exist two positive integers $m \in A$ and $n \notin A$ each of which is a product of k distinct elements of S for some $k \geq 2$.

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